

Supplementary Information

for “HPC-FLRNet: A Compact Deep Learning Model for Compressive Strength Prediction of High-Performance Concrete under Small Sample Conditions”

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Supplementary Note S1. HPC1826 construction and robust quality screening

This note provides the detailed, reproducible procedure used to construct the HPC1826 benchmark from three public sources, including source-field inspection, feature and unit harmonization, deduplication, and robust quality screening.

S1.1 Source datasets and harmonization

Three publicly available datasets were consolidated into a unified feature space. Feature names and physical units were aligned, and only common mix-design descriptors and curing age were retained. During harmonization, source-specific fields not required by the unified 10-feature schema were excluded. In particular, data103 had no additional fields removed; data714 excluded tensile strength of cement fct (MPa), splitting tensile strength fst,t (MPa), slump (mm), sand ratio (%), fineness modulus of sand, stone powder content in sand (%), and Dmax of crushed stone (mm); and data1133 excluded superplasticizer (kg in a m³ mixture) because the unified feature schema used SP/C as the aligned descriptor. After merging, the raw combined set contained 1,952 records. Exact duplicates were removed, yielding 1,925 unique records for subsequent screening.

For sources where ratio-type descriptors were not explicitly reported, W/B, W/C, and SP/C were calculated deterministically from the available input constituent quantities according to conventional concrete mix-design definitions. For records reported at a fixed 28-day curing age without an explicit age column, Age was set to 28 days. After alignment, the retained modeling variables contained no missing entries.

S1.2 Robust outlier screening (IQR-based; conservative intersection rule)

Because most mixture variables and strength values exhibit skewness and heavy tails, we adopted a non-parametric interquartile range (IQR) screening approach rather than normality-assumption-based z-score thresholds. For each numeric feature, records outside the standard IQR fences [$Q1 - 1.5 \cdot IQR$, $Q3 + 1.5 \cdot IQR$] were flagged. To avoid excessive removal of rare-but-physically plausible mixtures, a conservative intersection rule was used: a record was removed only when it was consistently flagged across multiple variables rather than by a single-feature trigger.

S1.3 Accounting of sample counts

After screening, 99 records (5.14%) were removed from the intermediate set (1,925 $\rightarrow$ 1,826), producing the final HPC1826 benchmark.

Supplementary Note S2. Cross-source distribution overlap analysis

To evidence cross-source heterogeneity, we provide a source-aware PCA projection of the merged data. The visualization indicates partial overlap among sources alongside source-dependent shifts, motivating robust evaluation across multiple datasets and cautious interpretation when extrapolating beyond the pooled benchmark distribution.

Supplementary Note S3. Baseline reproduction and configuration strategy

All baseline performance values reported in the manuscript were reproduced by the authors under a unified evaluation framework. Literature-reported configurations were used only as reference hyperparameter settings for defining reasonable model configurations; no

baseline performance values were directly copied from previous publications. Baselines were tuned using validation performance while keeping the held-out test set untouched. The tuning strategy and candidate ranges are summarized in Table S3. The final hyperparameter configurations used for the representative single-split results are reported in Tables 3-6 of the main manuscript.

Supplementary Note S4. Additional interpretability visualizations (standardized dependence plots)

We provide additional SHAP dependence plots with each x-axis standardized as a z-score. Standardization improves comparability across features with different physical units and reduces visual bias caused by scale differences. Because several mix-design descriptors are correlated, these SHAP plots should be interpreted as conditional feature contributions under dependency rather than as strict causal effects.

Supplementary Note S5. Statistical testing details and post-hoc outputs

Repeated-run experiments were conducted with 10 runs per model per dataset. The run index was treated as the blocking factor. We used the Friedman test to assess overall differences across methods, followed by paired Wilcoxon signed-rank post-hoc comparisons with Holm correction. The summary statistics are reported in Table S4, and ablation post-hoc statistics are reported in Table S5. Full CSV outputs are provided as supplementary data files in the released repository.

Supplementary Note S6. Five-fold cross-validation robustness check

To assess split sensitivity under small-sample conditions, we conducted a five-fold cross-validation robustness check for HPC-FLRNet on data103, data714, data1133, and HPC1826. In each fold, one subset was held out for testing, while 10% of the remaining training portion was used as an internal validation subset for early stopping and learning-rate scheduling. Feature scaling parameters were fitted only on the training subset within each fold and then applied to the corresponding validation and test subsets. The cross-validation results are summarized in Supplementary Table S6.

Supplementary Tables

Table S1. HPC1826 construction accounting.

Stage	Records	Description
Merged raw records	1952	After feature/unit alignment and merge of three public sources
After deduplication	1925	Exact duplicate rows removed
Final HPC1826	1826	After robust quality screening using the IQR-based conservative intersection rule

Table S2. Input features in HPC1826 (aligned names and units).

Feature	Symbol (paper)	Unit
Cement content	Cem	kg/m ³
Slag content	Slg	kg/m ³
Fly ash content	FA	kg/m ³
Water content	Wat	kg/m ³
Water-to-binder ratio	W/B	ratio
Water-to-cement ratio	W/C	ratio
Superplasticizer-to-cement ratio	SP/C	ratio
Coarse aggregate content	CA	kg/m ³
Fine aggregate content	FAgg	kg/m ³
Curing age	Age	days

Table S3. Baseline tuning ranges (candidate grids used in scripts; selected settings are reported in the main manuscript).

Model	Key tuned hyperparameters	Candidate values (grid)
SVR (RBF)	C, epsilon, gamma	$C \in \{5, 10, 20, 50, 100, 200, 500, 1000\}$; $\epsilon \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09\}$; $\gamma \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$
GBRT	n_estimators, max_depth, learning_rate	$n_estimators \in \{100, 200, 300, 500, 1000\}$; $max_depth \in \{1, 2, 3, 4, 5, 6\}$; $learning_rate \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$
XGBoost	n_estimators, max_depth, learning_rate	$n_estimators \in \{100, 200, 300, 500, 1000\}$; $max_depth \in \{1, 2, 3, 4, 5, 6\}$; $learning_rate \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$
MLP	hidden_dim1, hidden_dim2, max_iter	$hidden_dim1 \in \{100, 200, 300, 400, 500\}$; $hidden_dim2 \in \{100, 200, 300, 400, 500\}$; $max_iter \in \{100, 200, 300, 500, 1000\}$

Table S4. Friedman test summary and average ranks (RMSE; lower is better).

Dataset	Friedman χ^2	p-value	Best/top ranks (average rank)
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data103	36.64	2.14×10^{-7}	HPC-FLRNet (1.0) < MLP (2.3) < SVR (2.7)
data714	26.96	2.03×10^{-5}	GBRT (2.0) = XGBoost (2.0) $\approx$ HPC-FLRNet (2.2)
data1133	29.20	7.12×10^{-6}	HPC-FLRNet (1.7) $\approx$ XGBoost (2.0) < GBRT (2.4)
HPC1826	33.04	1.17×10^{-6}	HPC-FLRNet (1.7) < GBRT (1.9) < XGBoost (2.4)

Table S5. Ablation planned post-hoc comparisons (Full vs each variant; RMSE; Holm correction $m = 5$).

Variant	Δ RMSE vs Full	% vs Full	p_raw	p_Holm
without CRM	0.8283	20.74%	0.00506	0.02531
without FRM	0.3444	8.62%	0.00506	0.02531
without IPM	0.2079	5.21%	0.00506	0.02531
without LayerNorm	0.1644	4.12%	0.00506	0.02531
without Dropout	0.2464	6.17%	0.00691	0.02531

Table S6. Five-fold cross-validation results of HPC-FLRNet on four datasets.

Panel A. Summary statistics over five folds (mean $\pm$ standard deviation).

Dataset	R ²	R	RMSE	MAE	MAPE
data103	0.979 $\pm$ 0.012	0.989 $\pm$ 0.006	1.038 $\pm$ 0.269	0.831 $\pm$ 0.231	2.470 $\pm$ 0.795
data714	0.955 $\pm$ 0.015	0.977 $\pm$ 0.008	4.562 $\pm$ 0.750	3.174 $\pm$ 0.455	10.517 $\pm$ 2.914
data1133	0.913 $\pm$ 0.012	0.955 $\pm$ 0.006	4.729 $\pm$ 0.226	3.306 $\pm$ 0.261	11.274 $\pm$ 1.808
HPC1826	0.935 $\pm$ 0.015	0.967 $\pm$ 0.008	4.746 $\pm$ 0.434	3.398 $\pm$ 0.208	10.775 $\pm$ 0.482

Panel B. Fold-wise metrics.

Dataset	Fold	n_train	n_val	n_test	R ²	R	RMSE	MAE	MAPE
data103	1	73	9	21	0.978	0.989	1.011	0.851	2.409
data103	2	73	9	21	0.986	0.993	1.150	0.874	2.628
data103	3	73	9	21	0.980	0.990	0.831	0.695	1.900
data103	4	74	9	20	0.960	0.980	1.434	1.175	3.722
data103	5	74	9	20	0.992	0.996	0.763	0.559	1.691
data714	1	513	58	143	0.951	0.975	4.848	3.546	11.016
data714	2	513	58	143	0.950	0.975	4.799	3.132	9.216
data714	3	513	58	143	0.941	0.970	5.146	3.587	15.193
data714	4	513	58	143	0.951	0.975	4.769	3.151	9.735
data714	5	514	58	142	0.981	0.990	3.248	2.456	7.426
data1133	1	815	91	227	0.923	0.961	4.523	2.971	9.657
data1133	2	815	91	227	0.926	0.962	4.477	3.195	11.325
data1133	3	815	91	227	0.897	0.947	5.013	3.409	10.313
data1133	4	816	91	226	0.911	0.954	4.780	3.277	10.756
data1133	5	816	91	226	0.907	0.952	4.854	3.677	14.318
HPC1826	1	1314	146	366	0.951	0.975	4.295	3.143	10.557
HPC1826	2	1314	147	365	0.934	0.967	4.936	3.580	11.234
HPC1826	3	1314	147	365	0.928	0.963	4.845	3.391	10.731
HPC1826	4	1314	147	365	0.913	0.956	5.323	3.628	11.248
HPC1826	5	1314	147	365	0.947	0.973	4.332	3.248	10.107

Note: In each fold, the held-out fold was used only for testing, while 10% of the remaining training portion was used as an internal validation subset for early stopping and learning-rate scheduling. Feature scaling parameters were fitted only on the training subset within each fold.

Table S7. Split-wise performance of HPC-FLRNet on HPC1826 under the fixed split diagnostic.

Split	RMSE (mean $\pm$ standard deviation)	MAE (mean $\pm$ standard deviation)	R ² (mean $\pm$ standard deviation)
Train	3.046 $\pm$ 0.226	2.192 $\pm$ 0.190	0.9725 $\pm$ 0.0041
Validation	4.195 $\pm$ 0.127	3.041 $\pm$ 0.065	0.9471 $\pm$ 0.0032
Test	4.388 $\pm$ 0.094	3.159 $\pm$ 0.101	0.9409 $\pm$ 0.0025

Note: Metrics are computed on the training subset (excluding the validation subset), the held-out validation subset, and the test subset under the fixed split protocol. The fixed protocol uses 82% train and 18% test, with 10% of the training portion further held out as validation.

Supplementary Figures

Fig. S1. Source-aware PCA projection of HPC1826 (points colored by source) to evidence cross-source overlap and shifts.

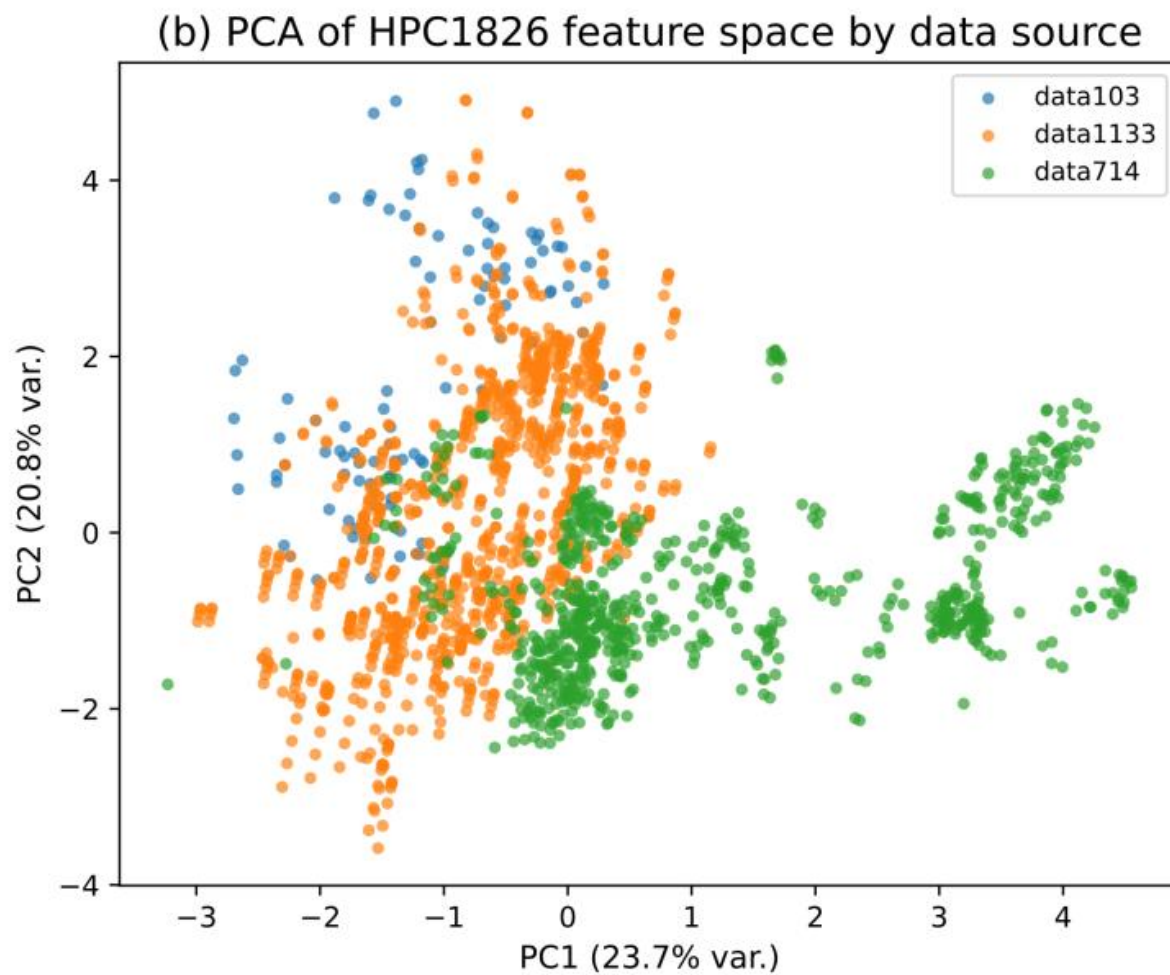


Fig. S2. Additional SHAP dependence plots (x-axis standardized as z-score) illustrating nonlinear effects and interactions.

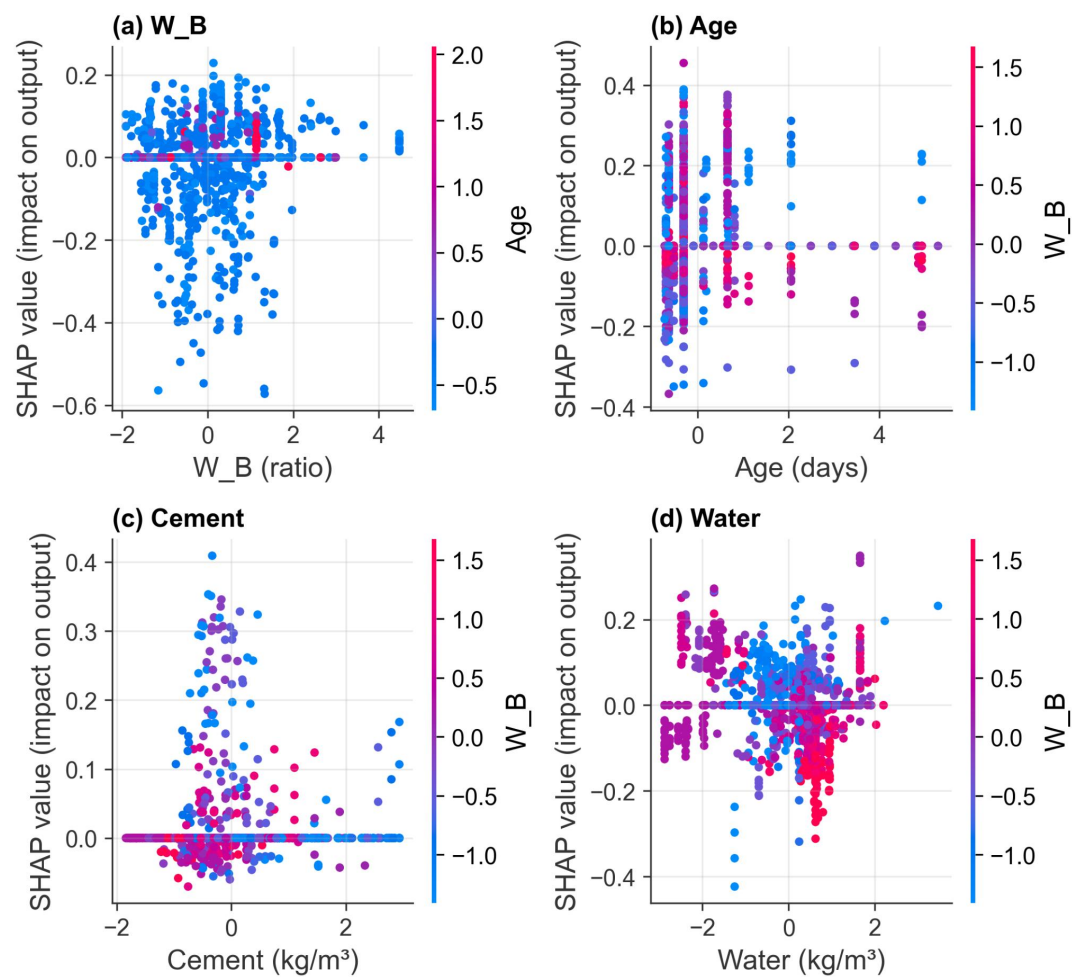


Fig. S3. SHAP global feature importance summary for HPC-FLRNet on HPC1826 (backup figure).

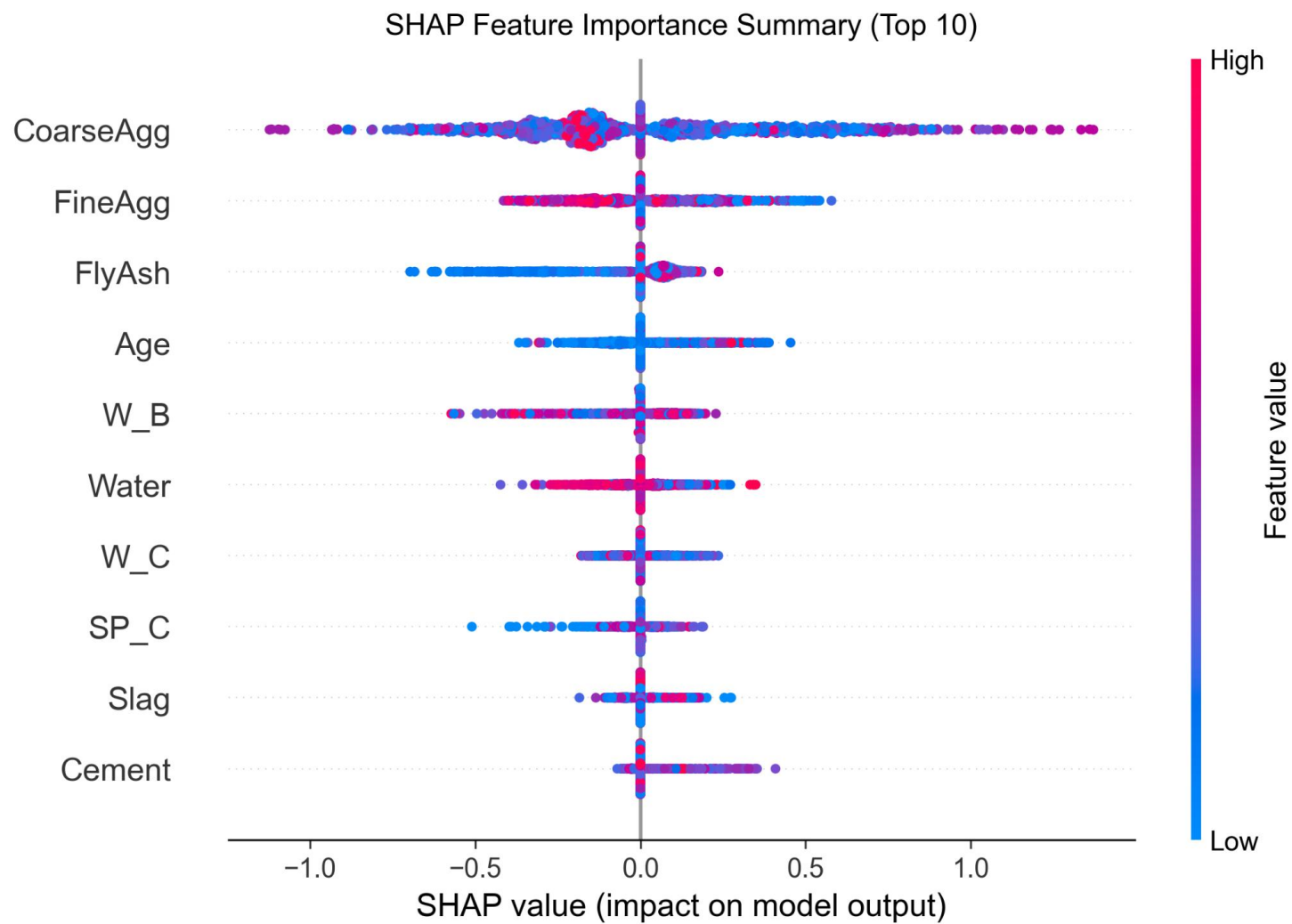


Fig. S4. Cross-check against classical water-binder empirical relations (Abrams/Bolomey-type trend) and model predictions.

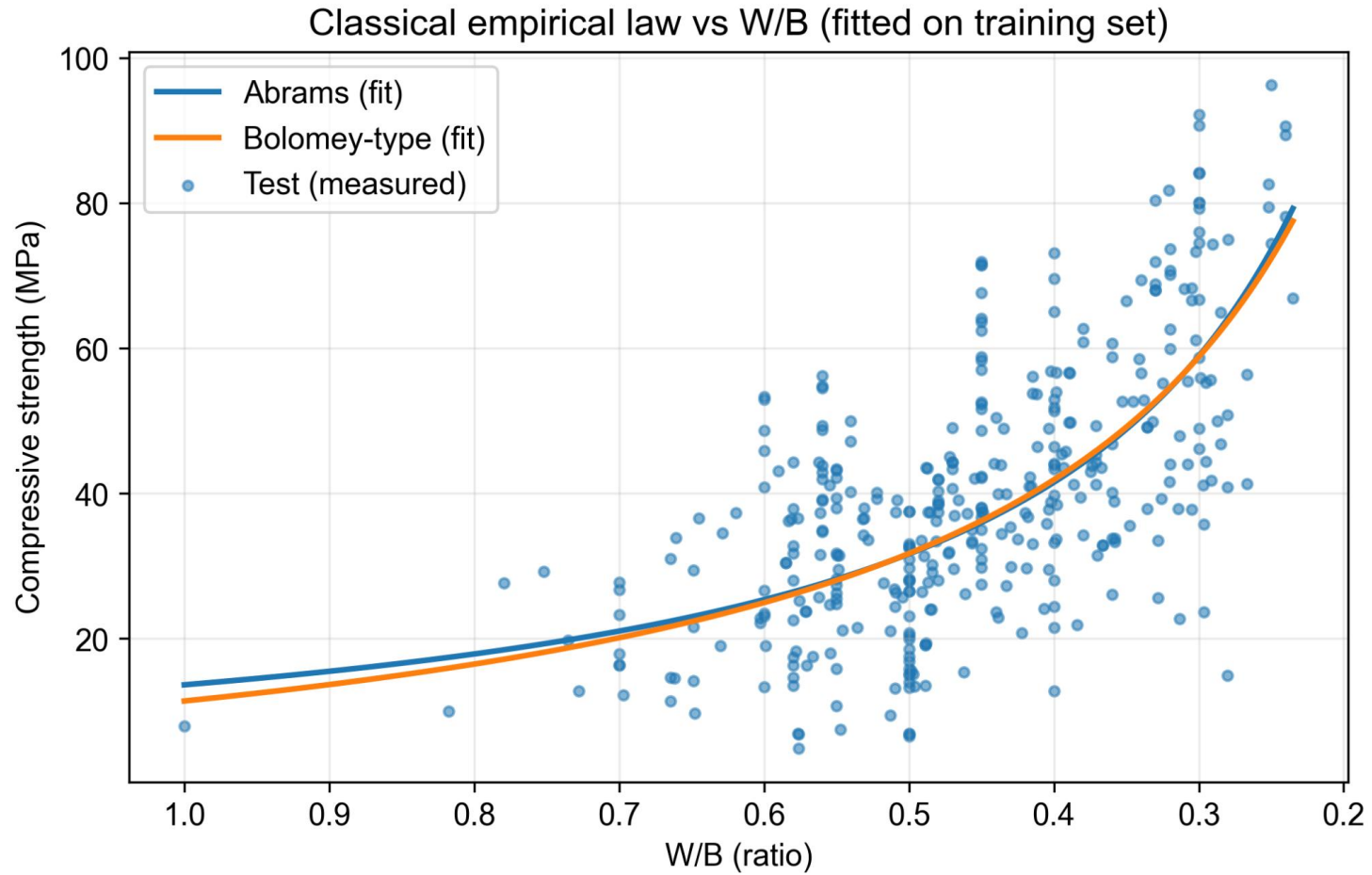
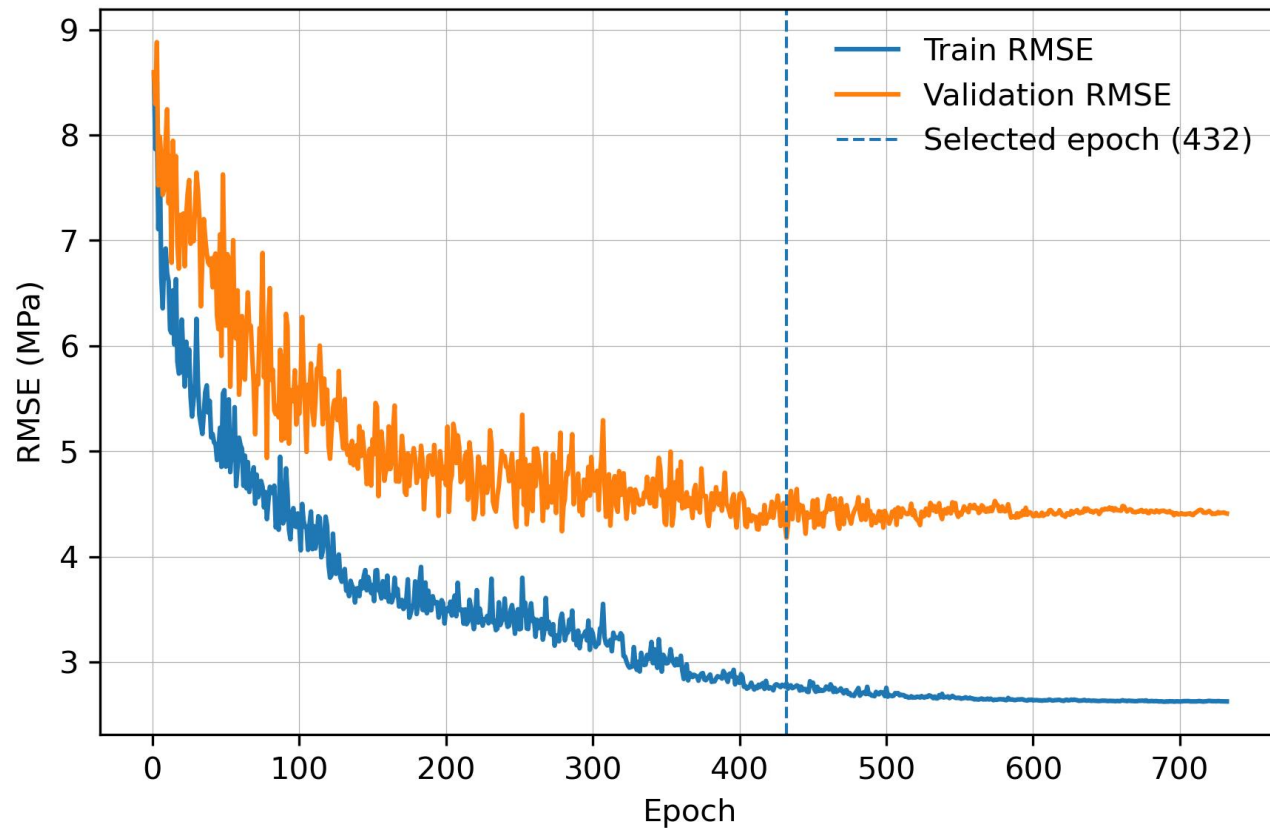


Fig. S5. Training and validation learning curves of HPC-FLRNet on HPC1826 under the fixed split. The selected checkpoint corresponds to the minimum validation loss (early_stop = 300).



Supplementary Data files

Supplementary Data 1 (data/intermediate/tmp_data1950_merged.xlsx): merged intermediate dataset after feature/unit alignment before deduplication and screening.

Supplementary Data 2 (data/intermediate/tmp_data1925_dedup.xlsx): intermediate dataset after deduplication.

Supplementary Data 3 (data/processed/HPC1826.xlsx): final benchmark dataset (HPC1826) after robust quality screening.

Supplementary Data 4-7 (results/significance/<dataset>/... and *results.json): statistical testing outputs for each dataset, including Friedman summaries, average ranks, and Wilcoxon-Holm post-hoc outputs.

Supplementary Data 8-11 (results-CV5/<dataset>/...): five-fold cross-validation fold metrics and summary files for data103, data714, data1133, and HPC1826.

All supplementary data files and scripts are available in the released repository: <https://github.com/tilerons/HPC-FLRNet>